Huawei Access Node QoS Congestion Event Investigation and Resolution SOP (KOR-ACC-83)

Node: KOR-ACC-83 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_3481

Description

The failure of Node ID KOR-ACC-83 at the Koramangala Datacenter, attributed to a QoS congestion event (microbursts, interface tail drops, QoS queue starvation), is a critical event that necessitates swift and precise resolution to prevent network instability and potential data losses. This SOP is designed to guide the investigation and resolution of such events on Huawei Access equipment, focusing on the root causes of QoS-related issues, specifically in the context of congestion events. The Huawei Access node's QoS functionality is based on the Differentiated Services (DiffServ) architecture, which employs Traffic Conditioning Agreements (TCAs) and Policers to regulate traffic and ensure Quality of Service (QoS) guarantees. However, in the event of microbursts or interface tail drops, the QoS queues can become congested, leading to packet drops and potential network instability. This SOP will guide the investigation of such events, identifying the root causes and implementing the necessary corrective actions to restore QoS guarantees and prevent future occurrences.

Pre-requisite Checks

- Verify Node ID KOR-ACC-83 is accessible via remote hands at the Koramangala Datacenter.
- Confirm redundant link status to ensure continued network connectivity during the investigation.
- Gather relevant logs and configuration files for further analysis.

Potential Root Causes

Cause	Probability
TCAs not properly configured or misconfigured, leading to QoS queue starvation.	High
Policer not configured or misconfigured, resulting in microbursts.	Medium
Interface congestion due to excessive traffic or incorrect interface configuration.	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos tca, Expected Output: TCAs configured on the node, including TCAs for each QoS class., Description: This command displays the configured TCAs on the node, including the Traffic Class, Action, and Action Argument. This information will help identify if TCAs are properly configured or if there are any misconfigurations that may be contributing to the QoS congestion event.

Command: display qos policer, Expected Output: Policers configured on the node, including policer parameters., Description: This command displays the configured policers on the node, including the policer name, policer type, and policer parameters. This information will help identify if policers are properly configured or if there are any
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```
misconfigurations that may be contributing to the microbursts or QoS
queue starvation.

Command: display interface congestion, Expected Output: Interface
congestion statistics, including packet drops and interface
utilization., Description: This command displays interface congestion
statistics, including packet drops and interface utilization. This
information will help identify if the interface is congested and if
there are any potential issues with the interface configuration or
excessive traffic.
```

Resolution Steps

Step 1: Step: Verify and adjust TCAs as necessary., Command: configure qos tca , Description: Verify the TCAs configured on the node and adjust them as necessary to ensure proper QoS guarantees. Use the `configure qos tca` command to modify the TCAs, specifying the TCA name and parameters as needed.

Step 2: Step: Verify and adjust policers as necessary., Command: configure qos policer , Description: Verify the policers configured on the node and adjust them as necessary to prevent microbursts and QoS queue starvation. Use the `configure qos policer` command to modify the policers, specifying the policer name and parameters as needed.

Step 3: Step: Verify and adjust interface configuration as necessary., Command: configure interface , Description: Verify the interface configuration and adjust it as necessary to prevent congestion and packet drops. Use the `configure interface` command to modify the interface configuration, specifying the interface name and parameters as needed.

Post-Check Verification

```
Command: display qos tca, Expected Output: TCAs properly configured
and without misconfigurations., Description: Verify that TCAs are
properly configured and without misconfigurations to ensure QoS
guarantees.

Command: display qos policer, Expected Output: Policers properly
configured and without misconfigurations., Description: Verify that
policers are properly configured and without misconfigurations to
prevent microbursts and QoS queue starvation.

Command: display interface congestion, Expected Output: Interface
congestion statistics showing improved performance., Description:
Verify that interface congestion statistics show improved performance,
indicating that the corrective actions have resolved the issue.
```

Roll-back Procedures

- Step: Restore TCAs to their original configuration., Command: undo configure qos tca , Description: Restore the TCAs to their original configuration if the corrective actions did not resolve the issue. Use the `undo configure qos tca` command to restore the original TCA configuration.

- Step: Restore policers to their original configuration., Command: undo configure qos policer , Description: Restore the policers to their original configuration if the corrective actions did not resolve the issue. Use the `undo configure qos policer` command to restore the original policer configuration.

- Step: Restore interface configuration to its original state., Command: undo configure interface , Description: Restore the interface configuration to its original state if the corrective actions did not resolve the issue. Use the `undo configure interface` command to restore the original interface configuration.